

SKILL WALLET

DATA ANALYTICS INTERNSHIP PROGRAM

PROJECT DOCUMENTATION

Visualizing Housing Market Trends

An Analysis of Sale Prices and Features using Tableau

INTERNSHIP MODULE

Dashboard Development and Data Visualization using Tableau

TEAM MEMBERS

Siddhi Mishra
Amarjeet Mishra
Mohit Tripathi
Anmol Kumar
Shikha Sharma

TEAM SIZE

5

TEAM ID

SWTID-2026-9687

INTERNSHIP PROVIDER

Skill Wallet

SUBMISSION DATE

9th July 2026

1 Objective

The objective of this project is to analyze a residential housing sales dataset and to design an interactive Tableau dashboard that visualizes the relationship between key property attributes and Sale Price. The project focuses specifically on how three factors influence pricing: Overall Grade (construction and design quality), Flat Area (living space in square feet), and Number of Bedrooms — presenting these relationships through clear, interactive visualizations that support quick, data-driven decision-making for buyers, sellers, and real estate stakeholders.

The project also aims to demonstrate the complete data analytics workflow followed during the internship: collecting a raw dataset from a public source, cleaning and preparing it using Python, loading it into Tableau, and building a set of worksheets that are combined into a dashboard for interactive exploration.

2 Software Used

The following tools were used at different stages of the project:

Software / Library	Purpose
Tableau Desktop / Public (2026.2)	Building worksheets, dashboards, and interactive visualizations
Microsoft Excel	Initial inspection and handling of the CSV dataset
Python 3	Data cleaning and preprocessing
Pandas	Handling missing values, duplicates, column renaming, and exporting the cleaned dataset
NumPy	Supporting numerical operations during preprocessing

3 Dataset Collection

3.1 Dataset Source

The dataset used in this project was sourced from Kaggle, a public repository of open datasets. It is a residential property sales dataset containing house-level attributes (physical features, location coordinates, and condition/grade ratings) along with the corresponding sale price.

3.2 Download Process

- The dataset was downloaded from Kaggle in CSV format.
- The raw file was stored locally for preprocessing before being loaded into Tableau.

3.3 File Format

The dataset is a comma-separated values (.csv) file. The final, cleaned file connected to Tableau (Cleaned_Housing_Data.csv) uses UTF-8 character encoding with a comma field delimiter and a header row.

3.4 Dataset Verification

After cleaning, the dataset was verified to contain 21,598 records across 31 columns, with zero missing values and zero duplicate rows, confirming that it was ready for analysis and visualization.

3.5 Storage

The cleaned CSV file was stored locally on the analyst's system and connected to Tableau using a direct text-file connection (Tableau's "Text File" data source type).

4 Dataset Description

4.1 Dataset Size

- Number of rows (records): 21,598
- Number of columns (fields): 31

4.2 Column Description and Data Types

The table below lists every column in the cleaned dataset, its data type as read by Tableau, and a description of what it represents.

Column Name	Data Type	Description
Sale_Price	Float	Sale price of the house (target variable)
No of Bedrooms	Integer	Number of bedrooms in the house
No of Bathrooms	Float	Number of bathrooms (fractional values indicate half-baths)
Flat Area (in Sqft)	Float	Interior living area of the flat/house in square feet
Lot Area (in Sqft)	Float	Total land/lot area in square feet
No of Floors	Float	Number of floors/storeys in the house
No of Times Visited	Integer	Number of times the property was viewed by prospective buyers
Overall Grade	Integer	Overall construction and design quality rating (1-10 scale)
Area of the House from Basement (in Sqft)	Float	Above-basement (non-basement) area of the house in square feet
Basement Area (in Sqft)	Integer	Basement area of the house in square feet
Age of House (in Years)	Integer	Age of the property in years at time of sale
Latitude	Float	Geographic latitude of the property

Column Name	Data Type	Description
Longitude	Float	Geographic longitude of the property
Living Area after Renovation (in Sqft)	Float	Living area after renovation (if applicable)
Lot Area after Renovation (in Sqft)	Integer	Lot area after renovation (if applicable)
Years Since Renovation	Integer	Number of years since the property was last renovated (0 if never renovated)
Condition_of_the_House_Excellent	Integer (0/1)	One-hot indicator: house condition rated Excellent
Condition_of_the_House_Fair	Integer (0/1)	One-hot indicator: house condition rated Fair
Condition_of_the_House_Good	Integer (0/1)	One-hot indicator: house condition rated Good
Condition_of_the_House_Okay	Integer (0/1)	One-hot indicator: house condition rated Okay
Ever_Renovated_Yes	Integer (0/1)	One-hot indicator: whether the house was ever renovated
Waterfront_View_Yes	Integer (0/1)	One-hot indicator: whether the property has a waterfront view
Zipcode_Group_Zipcode_Group_1 to 9	Integer (0/1)	One-hot indicators for the zipcode group/cluster the property belongs to (9 groups)

4.3 Sample Records

A sample of the first three records from the cleaned dataset is shown below (a subset of columns is displayed for readability):

Sale_Price	Bedrooms	Bathrooms	Flat Area (Sqft)	Grade	Age (yrs)
221900	3	1.00	1180	7	63
538000	3	2.25	2570	7	67
180000	2	1.00	770	6	85

4.4 Data Quality Summary

- Missing values after cleaning: 0
- Duplicate rows after cleaning: 0
- Sale Price range: ₹75,000 to ₹1,129,575 (mean ≈ ₹511,451; median ≈ ₹450,000)
- Overall Grade range: 1 to 10
- Age of House range: 3 to 118 years

5 Loading Dataset into Tableau

5.1 Connecting the CSV File

The cleaned CSV file (Cleaned_Housing_Data.csv) was connected to Tableau using the “Text File” connector. Tableau read the file with a comma field delimiter, UTF-8 character encoding, and the first row used as the column header.

5.2 Checking Column Names and Data Types

On the Data Source page, all columns were verified against the cleaned CSV. Tableau automatically assigned data types on connection: numeric fields such as Sale_Price, Flat Area (in Sqft), and Overall Grade were read as Number (Decimal) or Number (Whole), while the one-hot encoded indicator columns (Condition_of_the_House_*, Ever_Renovated_Yes, Waterfront_View_Yes, Zipcode_Group_*) were also read as whole numbers (0/1) rather than as text categories.

5.3 Data Source Page Review and Dataset Verification

The cleaned housing dataset was successfully loaded into Tableau for analysis. The View Data preview confirms that 10,000 rows and 32 fields were available and readable at the point of inspection — the additional 32nd field (Sale Price (bin)) is a calculated field created inside Tableau by binning Sale_Price, in addition to the 31 original columns from the cleaned CSV. (Note: Tableau's View Data window displays a maximum of 10,000 rows per preview; the complete dataset connected to Tableau contains 21,598 records in total, as detailed in Section 4.2.) The successful loading of the dataset ensured that all required data could be used for creating worksheets, dashboards, and storyboards without any loading issues.

Cleaned_Housing_Data.csv Sale Price (bin)	Cleaned_Housing_Data.csv Zipcode Group Zipcode Gr	Cleaned_Housing_Data.csv Zipcode Group Zipcode Gr	Cleaned_Housing_Data.csv Zipcode Group Zipcode Gr	Cleaned_Housing_Data.csv Zipcode Group Zipcode Gr	Cleaned_Housing_Data.csv Zipcode Group Zipcode Gr	Cleaned_Housing_Data.csv Zipcode Group Zipcode Gr	Cleaned_Housing_Data.csv Zipcode Group Zipcode Gr
215,648	00000	00000	00000	00000	00000	00000	00000
512,164	00000	00001	00000	00000	00000	00000	00000
161,736	00000	00001	00000	00000	00000	00000	00000
593,032	00000	00000	00001	00000	00000	00000	00000
485,208	00000	00000	00000	00001	00000	00000	00000
1,105,196	00000	00000	00000	00000	00001	00000	00000
242,604	00000	00000	00000	00000	00000	00000	00000
269,560	00000	00000	00000	00000	00000	00000	00000
215,648	00001	00000	00000	00000	00000	00000	00000
296,516	00001	00000	00000	00000	00000	00000	00000
646,944	00000	00000	00000	00000	00001	00000	00000
458,252	00000	00000	00000	00000	00001	00000	00000
296,516	00000	00001	00000	00000	00000	00000	00000
377,384	00000	00000	00000	00000	00001	00000	00000
512,164	00000	00000	00001	00000	00000	00000	00000
646,944	00000	00001	00000	00000	00000	00000	00000
377,384	00000	00001	00000	00000	00000	00000	00000
458,252	00000	00000	00001	00000	00000	00000	00000
188,692	00000	00000	00000	00000	00000	00000	00000
215,648	00000	00000	00000	00000	00000	00000	00000
377,384	00001	00000	00000	00000	00000	00000	00000
1,105,196	00000	00000	00000	00000	00000	00000	00000

Figure 5.1: Cleaned_Housing_Data.csv previewed in Tableau's View Data window, confirming successful data load

5.4 Initial Exploration

- Latitude and Longitude were reviewed for potential use in a map visualization.
- Numeric fields that represent categories (e.g., the one-hot Condition and Zipcode Group columns) were noted for possible conversion to discrete/dimension roles where needed for specific charts.
- Sale_Price was confirmed as the primary measure of interest for all analysis.

6 Data Cleaning and Preprocessing

Preprocessing was performed in Python using the Pandas and NumPy libraries before the dataset was loaded into Tableau. The steps performed are described below.

6.1 Loading the Raw Dataset

Purpose: Load the raw Kaggle CSV file into a Pandas DataFrame for cleaning.

```
import pandas as pd

# Load dataset
df = pd.read_csv("housing_data.csv")
```

6.2 Handling Missing Values

Purpose: Remove records with missing/null values so that aggregations and visualizations in Tableau are not affected by incomplete rows.

Method: Rows containing null values were dropped using `dropna()`. (`fillna()` was also available as an option for imputing values where dropping was not appropriate.)

```
# Handle missing values
df = df.dropna()
```

Output: The cleaned dataset contains zero missing values across all 31 columns.

6.3 Removing Duplicate Records

Purpose: Ensure that repeated listings do not distort averages and counts in the dashboard.

```
# Remove duplicates  
df = df.drop_duplicates()
```

Output: The cleaned dataset contains zero duplicate rows.

6.4 Renaming Columns for Consistency

Purpose: Standardize column names by removing leading/trailing spaces, converting to lowercase, and replacing spaces with underscores, so that field names are consistent and easier to reference in code.

```
# Rename columns for consistency  
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")
```

6.5 Outlier Detection

Purpose: Identify unusually high or low values (for example, in Sale Price or Bedrooms) that could distort chart scales and averages.

Method: Outliers were visually detected using boxplots of key numeric fields.

6.6 Feature Engineering

Purpose: Convert categorical attributes into a numeric, one-hot encoded format so they can be aggregated and used on the Color/Filter shelves in Tableau.

Categorical attributes converted to one-hot indicator columns include: Condition of the House (Excellent / Fair / Good / Okay), Ever Renovated (Yes), Waterfront View (Yes), and Zipcode Group (nine location-based groups). Each resulting column holds a value of 1 where the category applies and 0 otherwise.

6.7 Exporting the Cleaned Dataset

Purpose: Save the cleaned, transformed dataset as a new CSV file that Tableau connects to, keeping the raw and cleaned files separate.

```
# Save cleaned dataset  
df.to_csv("housing_data_cleaned.csv", index=False)
```

Output: A cleaned file of 21,598 records and 31 columns, with no missing values or duplicates, connected to Tableau as Cleaned_Housing_Data.csv.

7 Dashboard Development

7.1 Dashboard Objective

The dashboard, titled “Housing Market Analysis Dashboard,” was built to bring together three complementary views of the housing dataset — the effect of construction quality (Overall Grade), the effect of living area (Flat Area), and the effect of Number of Bedrooms — on Sale Price, so that a viewer can quickly understand what drives price in this market from a single screen.

7.2 Design Planning

- Each worksheet (Avg Sale Price by Overall Grade, Flat Area vs Sale Price, Average Sale Price by Number of Bedrooms) was built and verified individually before being placed on the dashboard.
- Fields were chosen based on their relationship with Sale Price: Overall Grade and Flat Area both show a strong positive relationship with price, while Number of Bedrooms was included as an intuitive, buyer-facing feature.

7.3 Layout Selection

A tiled, three-column layout was used so that all three charts are visible side by side without scrolling: Avg Sale Price by Overall Grade on the left, Flat Area vs Sale Price in the center, and Average Sale Price by Number of Bedrooms on the right. The dashboard was sized for a Generic Desktop (1366 x 768) display, with a Device Preview layout also configured for Phone.

7.4 Color Scheme

A consistent blue color scheme was used across all three worksheets for visual coherence. In the Flat Area vs Sale Price scatter plot, marks are additionally color-encoded by No of Bedrooms on a sequential blue scale (light to dark, ranging from 0 to 33 bedrooms) and sized by Avg. Sale Price, so that larger/darker circles indicate higher-priced, larger-bedroom properties.

7.5 Interactive Filters

A Number of Bedrooms filter and an Overall Grade filter were added to allow users to focus the dashboard on specific segments of the market (see Section 10 for filter testing results).

7.6 Legends, Actions, and Formatting

- A shared “Avg. Sale Price” size legend and a “No of Bedrooms” color legend are displayed on the right-hand side of the dashboard for the scatter plot.
- Axis titles, gridlines, and consistent K-formatted price axes (e.g., 200K, 400K, 600K) were used across all three charts for easy comparison.
- A dashboard title (“Housing Market Analysis Dashboard”) was added at the top of the dashboard for clear identification.

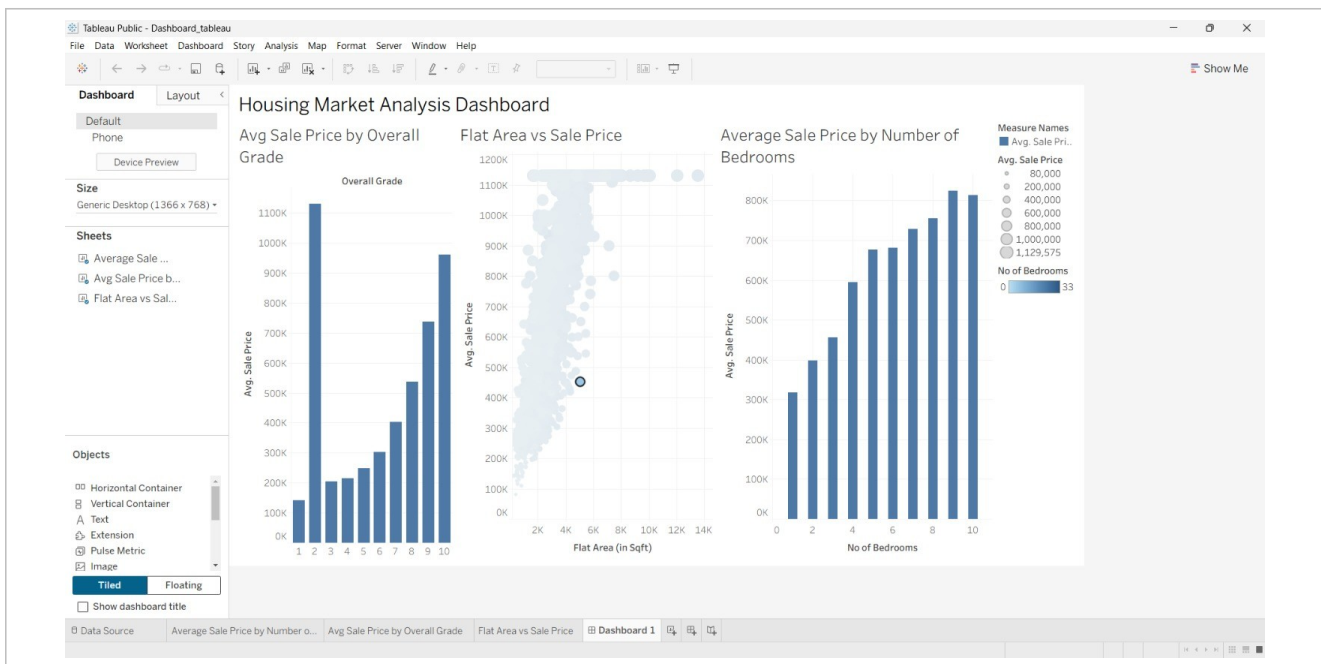


Figure 7.1: Complete Housing Market Analysis Dashboard in Tableau Public, combining all three worksheets

In addition to the dashboard, a Tableau Story ("Story 1") was built to walk viewers through each visualization one at a time, with a short narrative caption above each story point, before presenting the combined dashboard as the final story point.

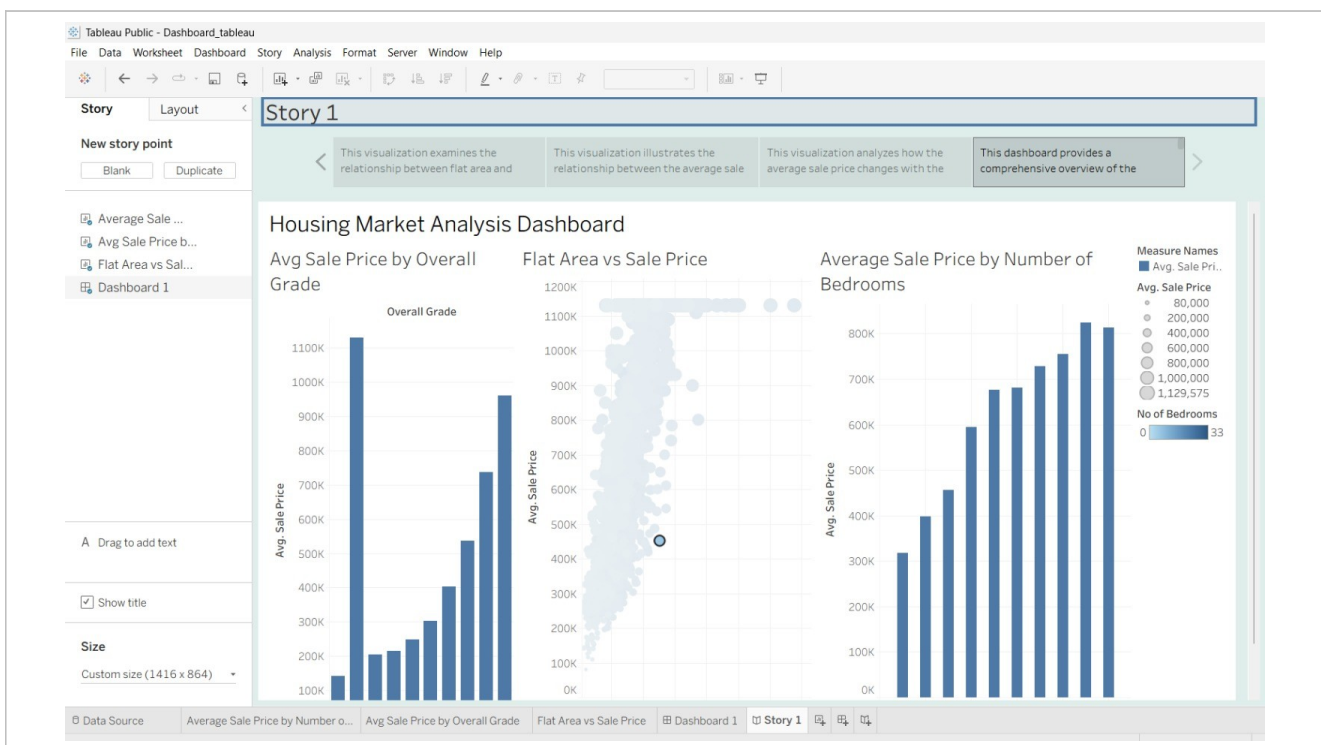


Figure 7.2: Final story point presenting the complete dashboard as a comprehensive overview

8 Visualization Documentation

This section documents each worksheet on the Housing Market Analysis Dashboard in detail, based on the actual field configuration and dashboard screenshots from the completed Tableau workbook.

8.1 Average Sale Price by Overall Grade

Objective

To analyze how the overall construction and design quality grade of a house affects its average selling price.

Chart Type

Bar Chart.

Fields Used

- Columns: Overall Grade
- Rows: Avg. Sale Price
- Filter: Overall Grade (all grades 1 through 10 included)

Steps to Create the Visualization

- Dragged Overall Grade to the Columns shelf.
- Dragged Sale_Price to the Rows shelf and set the aggregation to Average.
- Added Overall Grade to the Filters shelf and selected all grade values (1-10) to allow interactive control.
- Formatted the Y-axis to display values in K (thousands) for readability.

Why This Chart Was Selected

A bar chart is the clearest way to compare an averaged numeric value (Sale Price) across a small set of ordered categories (Grade 1 to 10).

Observation

Average sale price rises sharply and consistently with Overall Grade, from about ₹142,000 at Grade 1 to nearly ₹960,000 at Grade 10. Grade 2 shows an unusually high average ($\approx ₹1,129,575$) driven by a very small number of records at that grade, which stands out as an outlier against the otherwise steady upward trend from Grade 3 onward.

Business Insight

Overall Grade is one of the strongest, most consistent single predictors of price in this dataset. Sellers and agents can use the construction/design grade as a straightforward, defensible basis for setting price expectations with clients.

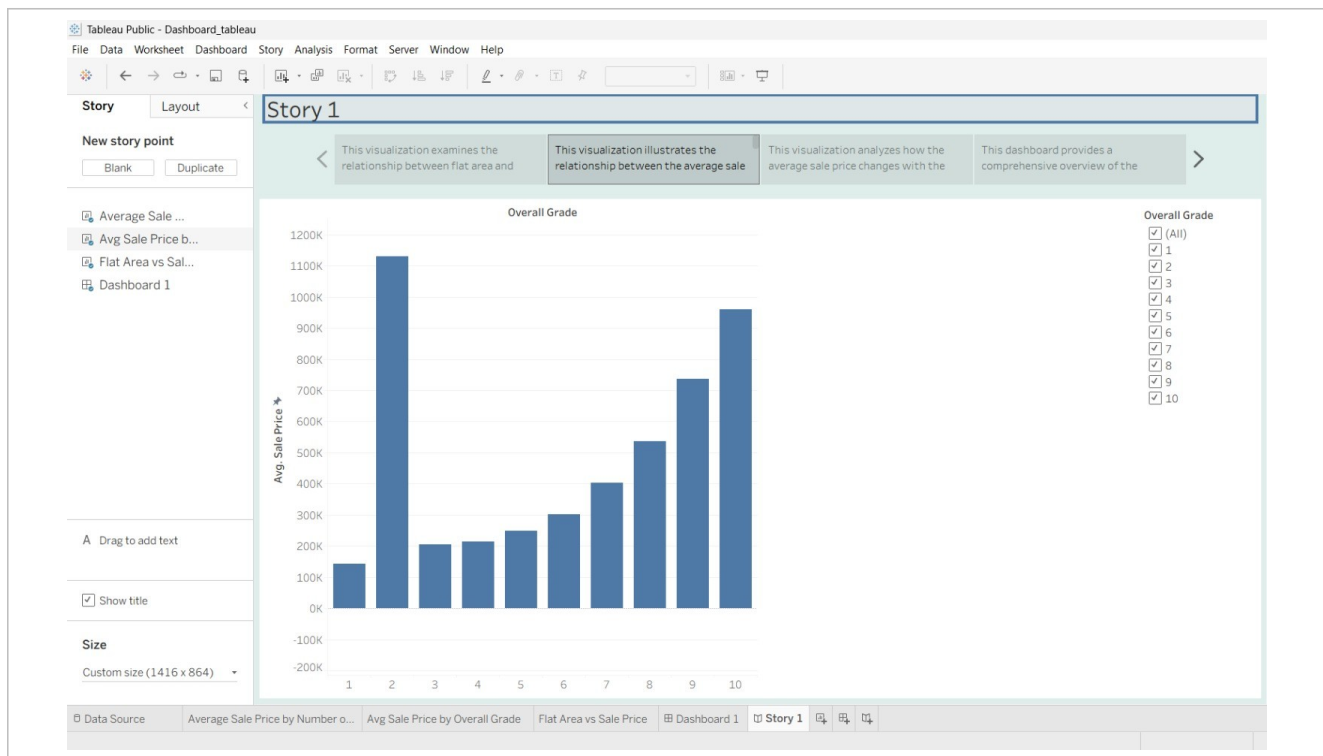


Figure 8.1: Average Sale Price by Overall Grade

8.2 Flat Area vs Sale Price

Objective

To identify the relationship between the interior living (flat) area of a house and its selling price.

Chart Type

Scatter Plot.

Fields Used

- Columns: Flat Area (in Sqft)
- Rows: Avg. Sale Price
- Size: Avg. Sale Price
- Color: No of Bedrooms

Steps to Create the Visualization

- Dragged Flat Area (in Sqft) to the Columns shelf and Sale_Price (as Average) to the Rows shelf.
- Changed the mark type to Circle.
- Dragged Avg. Sale Price to the Size shelf so that higher-priced points appear as larger circles.
- Dragged No of Bedrooms to the Color shelf, applying a sequential blue color scale (0 to 33 bedrooms).

Why This Chart Was Selected

A scatter plot is the standard way to visualize the relationship between two continuous numeric variables, and adding size and color encodings allows a third and fourth dimension (price magnitude and bedroom count) to be shown in the same view.

Observation

The scatter plot shows a clear positive relationship between flat area and sale price (correlation ≈ 0.69 in the underlying data): as flat area increases, average sale price generally increases as well. A small number of very large-area or very high-bedroom properties (including a data point with 33 bedrooms) appear as distinct outliers relative to the main cluster of points.

Business Insight

Flat area is a strong, intuitive driver of price, making it a useful field for quick price estimation. Buyers and sellers should be aware that a small number of outlier properties (unusually large area or bedroom count) can distort simple area-based price comparisons, and should be considered separately from typical listings.

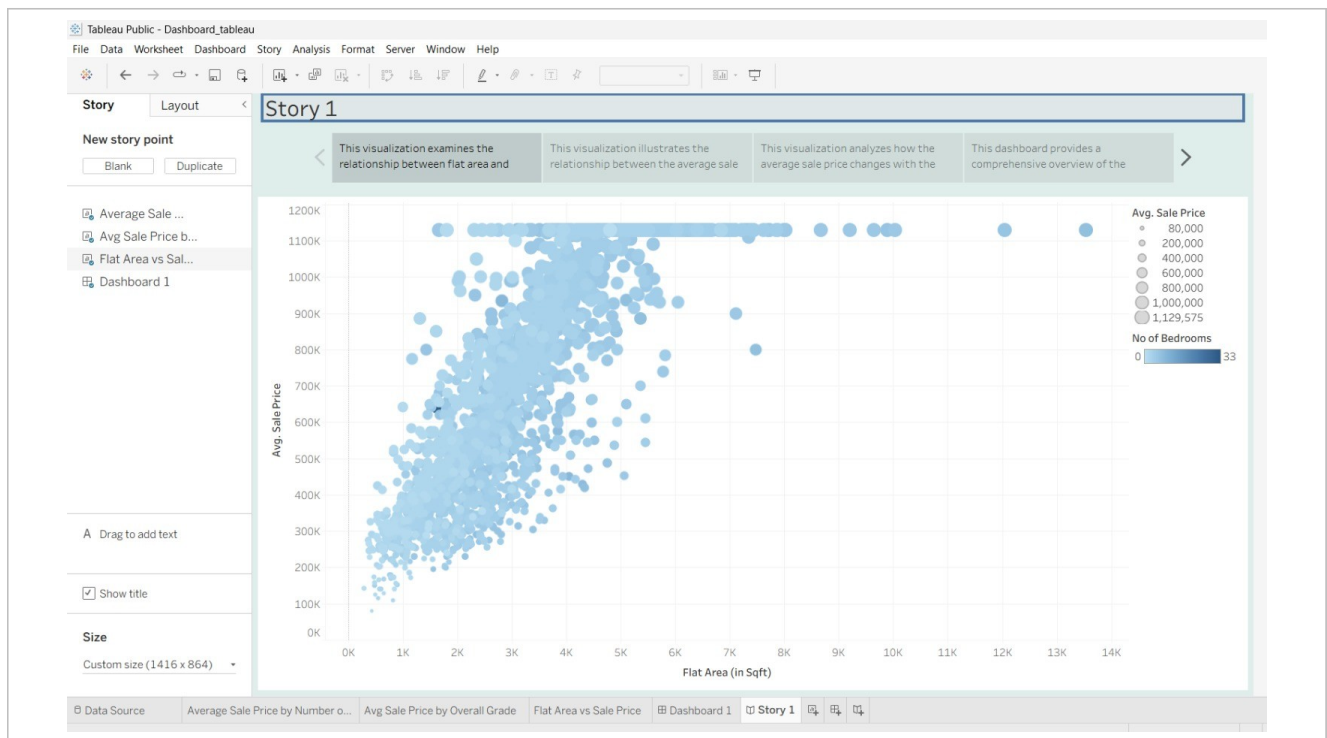


Figure 8.2: Flat Area vs Sale Price scatter plot, sized by Avg. Sale Price and colored by No of Bedrooms

8.3 Average Sale Price by Number of Bedrooms

Objective

To compare the average selling price of houses based on the number of bedrooms.

Chart Type

Bar Chart.

Fields Used

- Columns: No of Bedrooms
- Rows: Avg. Sale Price

Steps to Create the Visualization

- Dragged No of Bedrooms to the Columns shelf.
- Dragged Sale_Price to the Rows shelf and set the aggregation to Average.
- Formatted the Y-axis to display values in K (thousands) for consistency with the other worksheets.

Why This Chart Was Selected

A bar chart clearly shows how the average of a numeric measure (Sale Price) changes across an ordered category (Number of Bedrooms), matching the intuitive, buyer-facing nature of this field.

Observation

Average sale price increases steadily with the number of bedrooms, from about ₹317,000 for 1-bedroom homes to a peak of roughly ₹824,000 for 9-bedroom homes, before dipping slightly at 10 bedrooms (≈ ₹813,000) — likely reflecting the very small number of properties with that many bedrooms. Homes with 0 bedrooms (e.g., studio-style units) show a higher-than-expected average, again due to a very small sample size.

Business Insight

Bedroom count is a reliable, easy-to-communicate driver of price for the large majority of the market (roughly 1 to 7 bedrooms), where the trend is smooth and well-supported by many records. Real estate companies should treat the extreme ends of the range (0, 9, 10+ bedrooms) with caution when pricing, since these segments are based on very few properties.

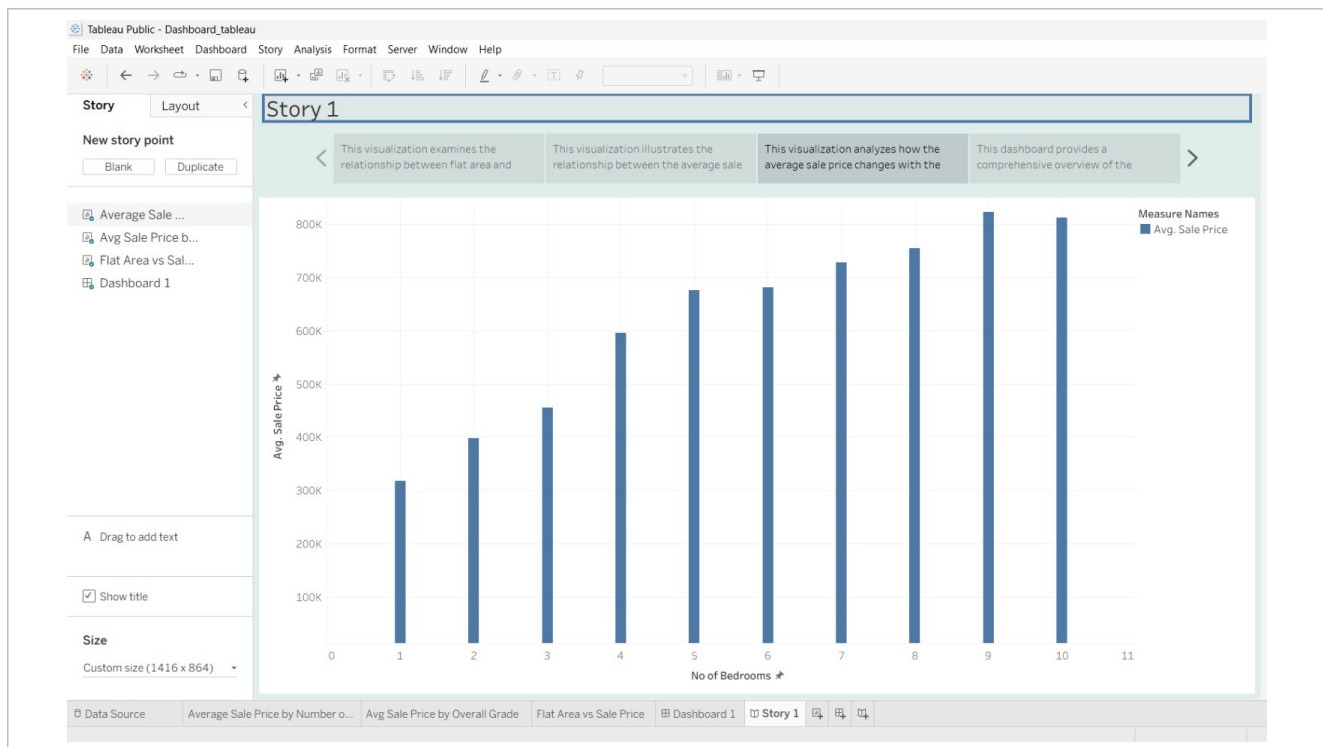


Figure 8.3: Average Sale Price by Number of Bedrooms

9 Dashboard Layout

Dashboard Title: Housing Market Analysis Dashboard

- Left: Avg Sale Price by Overall Grade (bar chart)
- Center: Flat Area vs Sale Price (scatter plot)
- Right: Average Sale Price by Number of Bedrooms (bar chart)
- Legends: Avg. Sale Price (size legend) and No of Bedrooms (color legend) are docked on the right-hand side of the dashboard, applying to the Flat Area vs Sale Price scatter plot.
- Filters: Number of Bedrooms and Overall Grade filters allow users to interactively narrow the view (see Section 10).
- Navigation: Users can hover over marks for tooltip detail, use the legends to highlight related marks across charts, and step through the accompanying Story for a guided narrative before reaching the full dashboard.

10 Interactive Filter Testing

A performance test was conducted to verify the functionality of interactive filters within the Tableau dashboard. Filters allow users to display specific data based on selected criteria, making the analysis more focused and user-friendly.

In this project, the Number of Bedrooms filter was applied to the dashboard. When a particular bedroom category was selected, all related visualizations updated dynamically to display only the relevant information. This confirmed that the filter was functioning correctly and improved the efficiency of data exploration.

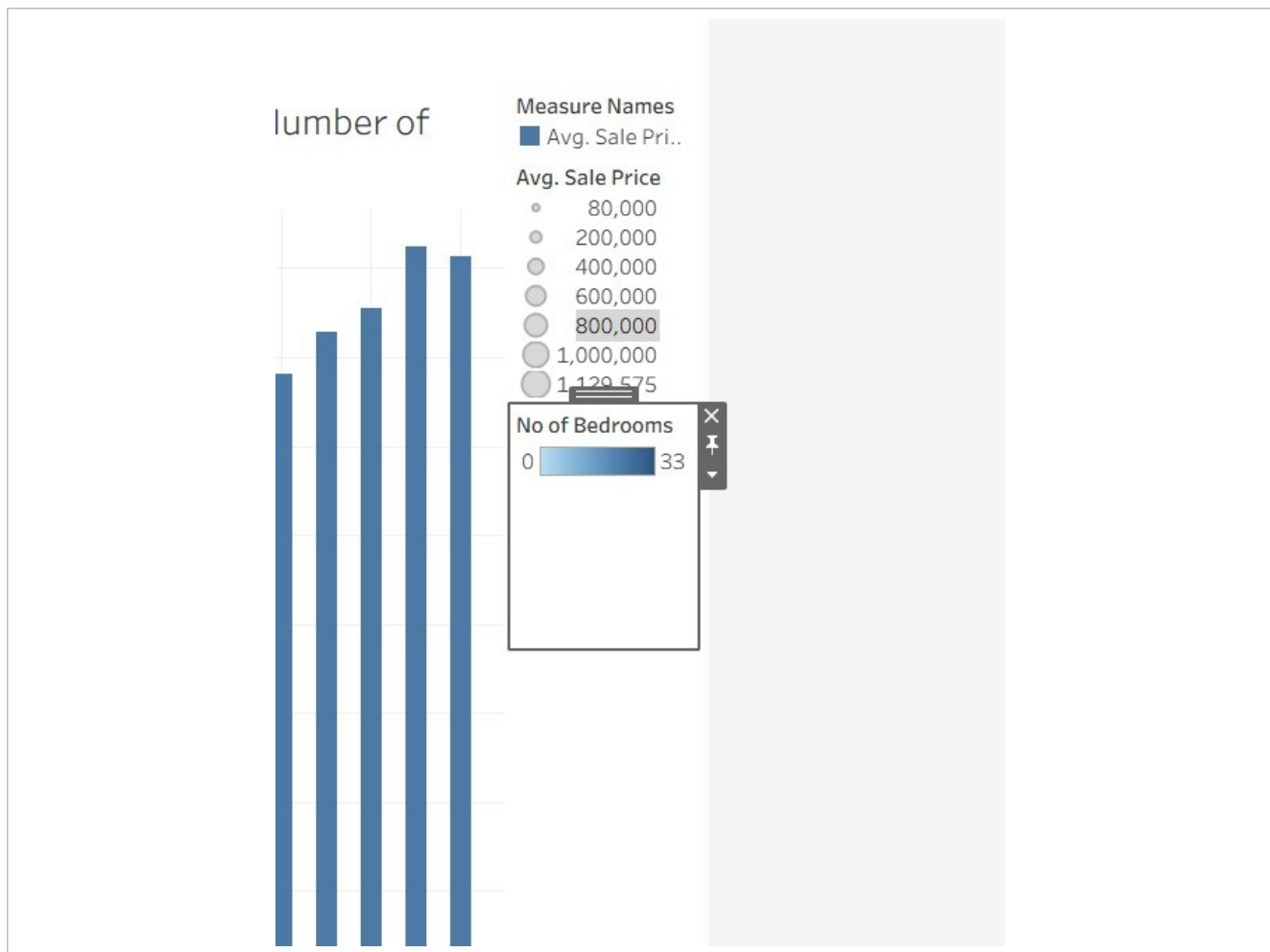


Figure 10.1: Number of Bedrooms filter and Avg. Sale Price legend applied on the dashboard

10.1 Test Result

Parameter	Result
Filter Used	Number of Bedrooms
Filter Response	Dynamic
Dashboard Update	Successful
Performance Status	Passed

10.2 Conclusion

The interactive filter successfully refined the housing market data based on the selected bedroom category. The dashboard responded immediately to user input, confirming that the filtering functionality worked efficiently.

11 Key Insights

- Average sale price rises sharply and consistently with Overall Grade, from about ₹142,000 at Grade 1 to nearly ₹960,000 at Grade 10.
- Flat Area shows a strong positive relationship with Sale Price (correlation ≈ 0.69); larger homes generally sell for more, though a small number of outlier properties exist.
- Average sale price increases steadily with the number of bedrooms across the bulk of the market (1 to 7 bedrooms), rising from $\approx$ ₹317,000 to $\approx$ ₹728,000.
- The scatter plot's bedroom-based color encoding reveals that most high-price, high-bedroom properties are concentrated in the upper range of flat area, reinforcing the link between size, bedrooms, and price.
- The dashboard's interactive Number of Bedrooms filter was tested and confirmed to update all related visualizations dynamically and successfully.
- A small number of extreme values (e.g., Grade 2's unusually high average, or the 33-bedroom property) act as outliers that should be interpreted separately from the main market trend.

12 Business Recommendations

12.1 For Buyers

- Use Overall Grade and Flat Area together as quick, reliable indicators of expected price range when comparing listings.
- Be cautious of listings at the extremes (very low or very high bedroom counts, or unusual grade/price combinations), as these are based on very few comparable properties in the market.

12.2 For Sellers

- Properties with a higher Overall Grade and larger Flat Area consistently command higher prices; sellers can use this relationship to set realistic, data-backed price expectations.
- Highlighting bedroom count in listings is worthwhile, since price rises steadily with bedrooms across the core of the market (1-7 bedrooms).

12.3 For Investors

- Since Grade and Flat Area are the strongest price drivers observed, investment or renovation decisions that improve either (e.g., quality upgrades or extensions) are likely to have the most direct impact on resale value.

12.4 For Real Estate Companies

- Interactive dashboards with filters (such as the Number of Bedrooms filter tested in Section 10) should be used in client-facing tools, since they allow prospective buyers to explore relevant listings quickly and confidently.
- Pricing tools should flag outlier segments (extreme bedroom counts, unusual grade/price combinations) so that sales teams do not rely on them for general market benchmarking.

13 Challenges Faced

- Ensuring the dataset had zero missing values and duplicates before connecting it to Tableau required careful verification after each cleaning step.
- Understanding Tableau's View Data preview limit (10,000 rows) versus the full connected dataset size required clarification to avoid misreporting the dataset's true size.
- Selecting appropriate size and color encodings for the Flat Area vs Sale Price scatter plot, given the wide range of bedroom values (0 to 33), required experimentation to keep the chart readable.
- Aligning the three worksheets consistently within the dashboard's tiled layout, while keeping legends and filters visible, required attention to Tableau's dashboard formatting options.
- Testing and confirming that the interactive filters updated all worksheets correctly added an additional validation step to the development process.

14 Learning Outcomes

- Gained hands-on experience connecting a cleaned CSV dataset to Tableau and configuring the Data Source page correctly.
- Learned to build multiple chart types in Tableau, including bar charts and scatter plots with combined size and color encodings.
- Practiced using Python (Pandas) for real-world data cleaning tasks: handling missing values, removing duplicates, renaming columns, and exporting cleaned data.
- Learned to build and test interactive dashboard filters, and to validate that dashboard interactivity works as intended.
- Gained experience building a Tableau Story to present visualizations in a guided, narrative sequence.
- Developed an understanding of dashboard design principles, including layout planning, use of filters, color encoding, and legends for interactivity.

15 Conclusion

This project demonstrates a complete data analytics workflow applied to a real-world housing dataset: sourcing the data from Kaggle, cleaning and preparing it using Python, loading it into Tableau, and building an interactive Housing Market Analysis Dashboard. The analysis shows that Overall Grade and Flat Area are the strongest drivers of Sale Price, and that Number of Bedrooms provides a clear, steady price signal across the core of the market. The dashboard's interactive filters were tested and confirmed to work reliably, giving users a fast, focused way to explore the data. Together, these visualizations provide a practical, at-a-glance view of the housing market that can support informed decisions by buyers, sellers, investors, and real estate companies.

16 Future Scope

- Predictive Analytics: Building a regression or machine learning model to predict Sale Price based on the available features.

- **Live Dashboards:** Connecting the dashboard to a live or regularly refreshed data source instead of a static CSV file.
- **Tableau Server / Cloud Publishing:** Publishing the dashboard to Tableau Server or Tableau Cloud for wider, controlled access by stakeholders.
- **Geospatial Analysis:** Using the Latitude/Longitude fields to build a map-based visualization of price trends by location and zipcode group.
- **Real-Time Data Integration:** Incorporating real-time listing data (e.g., via an API) to keep the dashboard current with the live market.